

Practical No: 07

Write a program to build a custom NER system.

```
In [ ]: print("125 Prosenjeet Shil")
```

125 Prosenjeet Shil

USING NLTK

```
In [ ]: import nltk
from nltk import word_tokenize, pos_tag, ne_chunk
nltk.download('maxent_ne_chunker_tab')

nltk.download('punkt')
nltk.download('maxent_ne_chunker')
nltk.download('words')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package maxent_ne_chunker_tab to
[nltk_data] /root/nltk_data...
[nltk_data] Package maxent_ne_chunker_tab is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data] /root/nltk_data...
[nltk_data] Package maxent_ne_chunker is already up-to-date!
[nltk_data] Downloading package words to /root/nltk_data...
[nltk_data] Package words is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
```

Out[]: True

```
In [ ]: nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
```

Out[]: True

```
In [ ]: #Sample text
text="I was born in Mumbai and went to Pune University for pursuing further stud

#Tokenizing the text
tokens = word_tokenize(text)

# Part of speech tagging
tagged_tokens=pos_tag(tokens)

#Named Entity Recognition (NE Chunking)
named_entities=ne_chunk(tagged_tokens)
```

```
#Print the Named Entities
print(named_entities)
```

```
(S
  I/PRP
  was/VBD
  born/VBN
  in/IN
  (GPE Mumbai/NNP)
  and/CC
  went/VBD
  to/TO
  (ORGANIZATION Pune/NNP University/NNP)
  for/IN
  pursuing/VBG
  further/JJ
  studies/NNS)
```

```
In [ ]: named_entities.draw()
```

```
-----
TclError                                Traceback (most recent call last)
<ipython-input-10-db8b50fdbf25> in <cell line: 0>()
----> 1 named_entities.draw()

/usr/local/lib/python3.11/dist-packages/nltk/tree/tree.py in draw(self)
    759         from nltk.draw.tree import draw_trees
    760
--> 761         draw_trees(self)
    762
    763     def pretty_print(self, sentence=None, highlight=(), stream=None, **kw
args):

/usr/local/lib/python3.11/dist-packages/nltk/draw/tree.py in draw_trees(*trees)
    1006         rtype: None
    1007         """
-> 1008         TreeView(*trees).mainloop()
    1009         return
    1010

/usr/local/lib/python3.11/dist-packages/nltk/draw/tree.py in __init__(self, *tree
s)
    857         self._trees = trees
    858
--> 859         self._top = Tk()
    860         self._top.title("NLTK")
    861         self._top.bind("<Control-x>", self.destroy)

/usr/lib/python3.11/tkinter/__init__.py in __init__(self, screenName, baseName, c
lassName, useTk, sync, use)
    2324             baseName = baseName + ext
    2325             interactive = False
-> 2326             self.tk = _tkinter.create(screenName, baseName, className, intera
ctive, wantobjects, useTk, sync, use)
    2327             if useTk:
    2328                 self._loadtk()

TclError: no display name and no $DISPLAY environment variable
```

SPACY

```
In [ ]: import spacy
        nlp = spacy.load("en_core_web_sm")
```

```
In [ ]: # Sample text
        text = "Jamshedji Tata founded Tata Steel on 26 August 1907 in India.He was very

        # Process the text using spaCy
        doc = nlp(text)

        # Print Named Entities
        for ent in doc.ents:
            print(ent.text, "->", ent.label_)
```

Jamshedji Tata -> PERSON
Tata Steel -> ORG
26 August 1907 -> DATE
India -> GPE
English -> LANGUAGE
the First World War -> EVENT
1914-1918 -> CARDINAL
the Tata Group -> ORG
Tata Motors -> ORG
800 -> MONEY
today -> DATE

```
In [ ]: # Visualizing Named Entities (Optional, works in Jupyter Notebook)
        spacy.displacy.render(doc, style="ent", jupyter=True)
```

Jamshedji Tata **PERSON** founded Tata Steel **ORG** on 26 August 1907 **DATE** in
India **GPE** .He was very fluent in English **LANGUAGE** .During the First World War
EVENT (1914-1918 **CARDINAL**), the Tata Group **ORG** made rapid progress. Tata
Motors **ORG** stock is worth ₹ 800 **MONEY** today **DATE** .